Do We Trust AI Because It Is Right or Because It Sounds Right?

Investigating Human Trust and Judgment of AI-Generated Information

Independent Research Project
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Artificial intelligence is increasingly used to provide information, solve problems, and support decision-making. However, AI-generated answers can appear convincing even when they are incorrect. This creates a significant challenge: whether people can distinguish reliable information from information that is simply presented confidently.

This study explores how the presentation of AI-generated answers, particularly differences between plain and polished formats, relates to people's trust and judgment of the information. An interactive experimental platform was developed using Python and Streamlit to present participants with ten AI-generated answers covering science, computer science, history, geography, and environmental topics. The answers varied in accuracy and presentation.

Participants evaluated whether each answer was correct, rated their confidence in their judgment, reported their level of trust in the answer, and indicated whether they would use the information. Responses were collected anonymously and analysed using Python.

The study focuses on trust calibration: whether a person's level of trust corresponds to the actual reliability of the information. The research is exploratory and uses a small sample of ten participants. The findings are intended to identify patterns in how people respond to AI-generated information and provide insight into the challenges of evaluating AI outputs.

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1. INTRODUCTION

Artificial intelligence has become increasingly common in education, research, work, and everyday life. People use AI systems to search for information, generate explanations, solve problems, write code, and support decisions. As these systems become more accessible, users are increasingly required to judge whether an AI-generated answer is reliable.

One difficulty is that AI-generated information can sound convincing even when it is incorrect. An answer may be written clearly, contain a detailed explanation, or be presented with a high level of confidence. These characteristics can influence how users perceive the answer, even though they do not necessarily indicate that the information is accurate.

This highlights an important distinction between trust and accuracy. An answer may appear reliable because it is presented convincingly, even when its content is incorrect. Similarly, a correct answer may receive less trust if it is presented with limited explanation.

This issue is closely related to trust calibration, which refers to how appropriately a person's trust corresponds to the actual reliability of a system or its outputs. Good calibration is important because users need to know when AI information can reasonably be relied upon and when it should be questioned or verified.

Previous research has examined human perceptions of AI accuracy, uncertainty communication, explanations, and reliance. However, these issues remain highly relevant as AI-generated information becomes part of everyday decision-making.

This project investigates whether the way an AI answer is presented affects how people judge and trust it. Rather than simply asking whether people trust AI, the study compares participants' perceptions with the actual accuracy of the answers.

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2. RESEARCH AIM AND QUESTIONS

2.1 Aim

The aim of this study is to investigate how presentation cues in AI-generated information influence people's trust and ability to judge whether an answer is reliable.

2.2 Main Research Question

How do confidence signals, explanations, and other presentation cues influence people's trust and judgment of AI-generated information?

2.3 Supporting Questions

1. Are participants more likely to trust AI-generated answers when they are presented with confidence signals or explanations?
2. Can participants accurately identify incorrect AI-generated answers?
3. Does confidence in a participant's judgment correspond to whether that judgment is correct?
4. Are participants more likely to trust an incorrect answer when it is presented in a convincing way?
5. What patterns of over-trust and under-trust can be observed?

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3. HYPOTHESIS

The study uses the following exploratory hypothesis:

Participants may be more likely to trust AI-generated answers when they are presented with confidence signals or supporting explanations, including cases where the answer itself is inaccurate.

The hypothesis explores a possible relationship rather than predicting a definite outcome. As not all presentation factors can be fully controlled, hence the study cannot determine whether one specific cue alone causes the observed effect

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4. Previous Research And Findings

4.1 Trust in AI

Trust is an important factor in human-AI interaction because users must decide whether an AI-generated output is reliable enough to accept or use.

Afroogh et al. (2024) reviewed research on trust in AI and highlighted the growing importance of understanding how people develop and use trust when interacting with increasingly capable AI systems.

However, trusting AI is not necessarily the same as using AI appropriately. Excessive trust can result in users accepting incorrect information, while insufficient trust can cause users to reject useful information.

This makes the relationship between trust and actual reliability particularly important.

4.2 Confidence and Human Judgment

AI systems can communicate confidence through language, numerical scores, or other signals. These signals may affect how users interpret an answer.

Steyvers et al. (2025) examined how people judge the accuracy of large language model responses and found differences between actual model performance and people's perceptions of model knowledge.

This suggests that users may not always accurately recognise whether an AI system is correct.

Yin (2025) also discussed the gap between machine confidence and human perceptions. Confidence therefore needs to be considered carefully because a confident presentation does not necessarily mean that an answer is correct.

4.3 Explanations and Trust

Explanations can make AI outputs easier to understand and may influence how users perceive their reliability.

Cheung and Ho (2025) examined the relationship between explainable AI and human trust. Their research demonstrates that explanations can affect human factors involved in trust.

However, an explanation does not automatically make an answer correct. An incorrect answer can also be accompanied by an explanation that makes it appear more convincing.

4.4 Uncertainty and Reliance

Uncertainty is another important factor in human-AI interaction.

Kim et al. (2024) investigated how uncertainty expressions from large language models affect user reliance and trust. Their work demonstrates that the way uncertainty is communicated can influence how users respond to AI outputs.

These findings are relevant to the current study because users may interpret confidence and uncertainty as signals of reliability.

4.5 Research Gap

Existing research has examined AI trust, confidence, explanations, uncertainty, and human assessment of AI accuracy.

This project combines these ideas within a small experimental study focused on the relationship between presentation cues, actual correctness, trust, and human judgment.

The purpose is to provide an independent investigation supported by a purpose-built software platform for collecting and analysing participant responses regarding this dilemma.

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5. METHODOLOGY

5.1 Research Design

The study uses a small-scale exploratory experimental design.

Participants evaluate ten AI-generated answers covering several factual topics. The answers include both correct and incorrect information and are presented using different combinations of presentation cues.

The experiment examines whether these differences are associated with changes in participant trust, confidence, and ability to identify incorrect information.

Because confidence, explanations, and correctness are not perfectly isolated across every question, the experiment is treated as exploratory rather than as a fully controlled factorial experiment.

5.2 Participants

The study uses a convenience sample of 30 participants.

Each participant evaluates ten questions, resulting in a maximum of:

30 participants × 10 questions = 300 individual responses.

The sample is intended as a pilot study. It is not large enough to represent the wider population, but it provides an initial dataset for examining patterns and evaluating the research design.

5.3 Experimental Materials

The experiment contains ten factual questions covering:

- Science
- Computer science
- History
- Geography
- Environmental issues

The AI-generated answers include both correct and incorrect information.

Different answers are presented with different combinations of information, including explanations and confidence signals.

5.4 Procedure

Participants complete the experiment through an interactive website.

For each question:

1. The participant reads the question and AI-generated answer.
2. The participant indicates whether they believe the answer is correct.
3. They rate their confidence in their judgment from 1–5.
4. They rate their trust in the AI answer from 1–5.
5. They indicate whether they would use the information.
6. They continue to the next question.

After all ten questions are completed, the responses are submitted and stored anonymously.

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6. SOFTWARE IMPLEMENTATION

A Python-based experimental platform was developed using Streamlit.

The platform was designed specifically for this research project rather than being a separate demonstration website. It allows the experiment to be conducted through a browser while automatically collecting participant responses.

The platform performs several functions:

- Generates anonymous participant IDs
- Displays experimental questions
- Presents AI-generated answers
- Displays presentation cues
- Collects participant judgments
- Records confidence ratings
- Records trust ratings
- Records intended reliance
- Stores responses for later analysis

Python was also used for data processing and visualisation through tools including Pandas and Matplotlib.

The software component therefore supports the full research process from participant interaction to data analysis.

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7. VARIABLES AND DATA ANALYSIS

The experiment records the following variables.

Actual Answer Correctness

Each answer has a known status:

- Correct
- Incorrect

This provides the objective reference for evaluating participant judgments.

Perceived Correctness

Participants classify each answer as:

- Correct
- Incorrect
- Unsure

This is used to calculate judgment accuracy.

Judgment Confidence

Participants rate their confidence from **1 to 5**.

Trust

Participants rate their trust in the AI answer from **1 to 5**.

Intended Reliance

Participants indicate:

- Yes
- No
- Unsure

when asked whether they would use the information.

Data Analysis

The analysis will primarily use descriptive statistics because of the small sample size.

Judgment accuracy will be calculated as:

Judgment Accuracy = Correct Judgments ÷ Total Judgments × 100

Average trust will be compared between correct and incorrect AI answers.

Participant confidence will also be compared with actual judgment accuracy.

The study will examine four broad trust-calibration patterns:

Actual Answer	Participant Trust	Interpretation
Correct	High	Appropriate trust
Correct	Low	Possible under-trust
Incorrect	High	Possible over-trust
Incorrect	Low	Appropriate scepticism

These categories will be used to identify overall patterns rather than to judge individual participants.

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8. RESULTS

A total of **10 participants** completed the study, producing **100 individual responses**.

Table 1. Summary of Results

Measure	Result
Number of participants	30
Questions per participant	10
Total responses	300
Overall judgment accuracy	60%
Average trust	3.19/5
Average confidence	3.29/5
Incorrect answers correctly identified	37.3%

8.1 Overall Judgment Accuracy

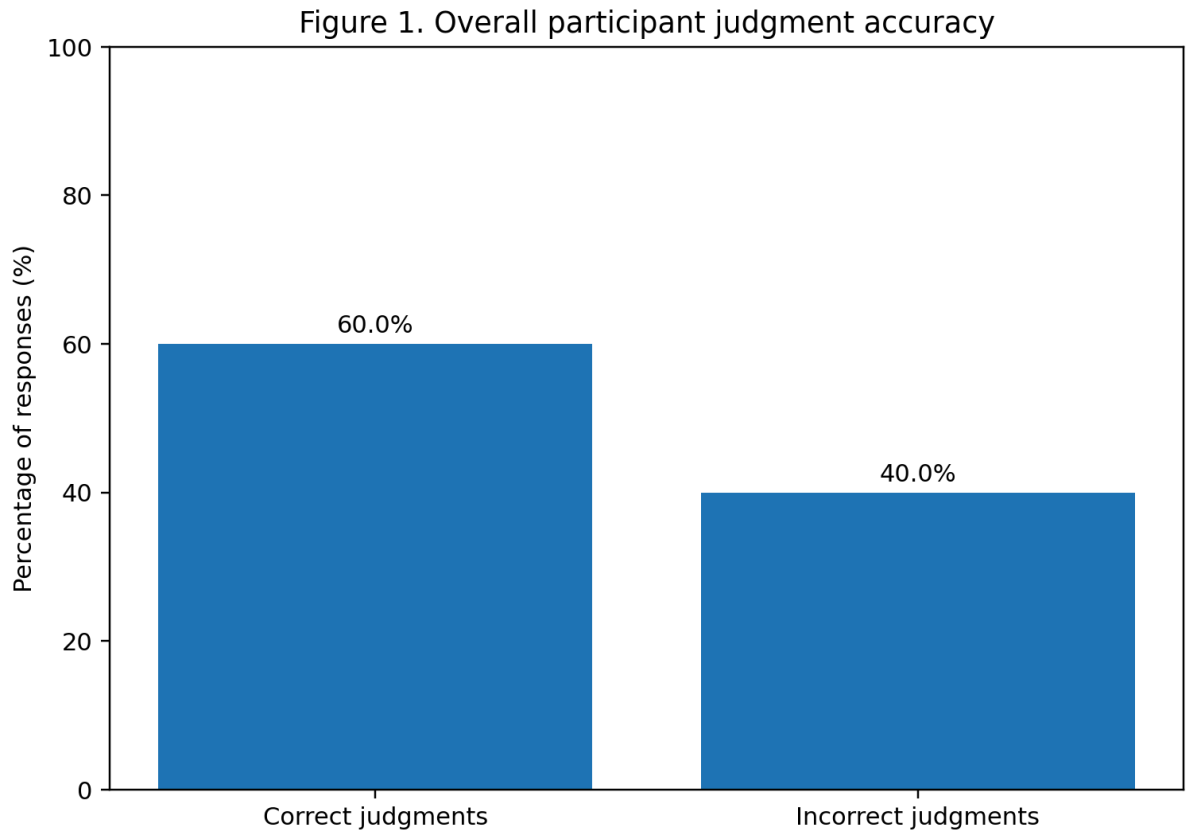


Figure 1. Overall participant judgment accuracy.

Participants correctly identified the status of **60%** of AI-generated answers.

This indicates that participants were generally able to distinguish between correct and incorrect AI-generated information.

8.2 Trust and Actual Accuracy

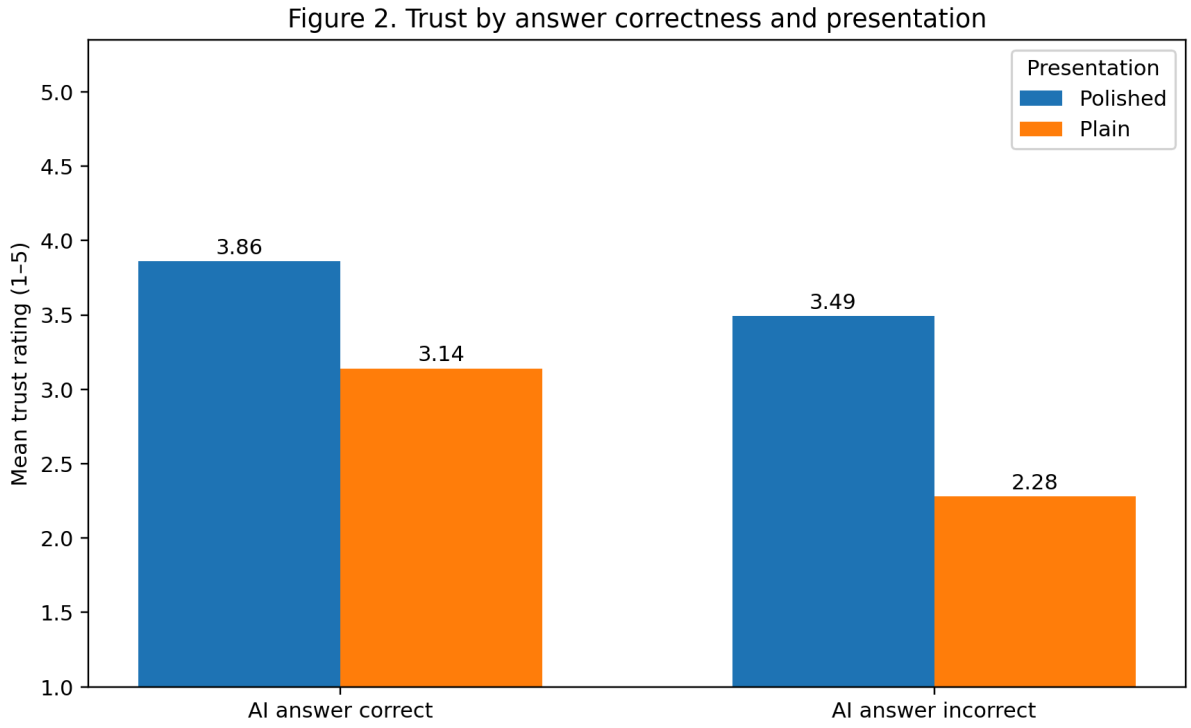


Figure 2. Average trust ratings for correct and incorrect AI-generated answers.

The average trust rating for correct answers was **3.52/5**, compared with **2.85/5** for incorrect answers.

The difference suggests that participant trust generally corresponded to actual answer accuracy. However, the distinction was not perfect, as some incorrect answers **still received relatively high trust ratings.**

8.3 Confidence and Judgment Accuracy

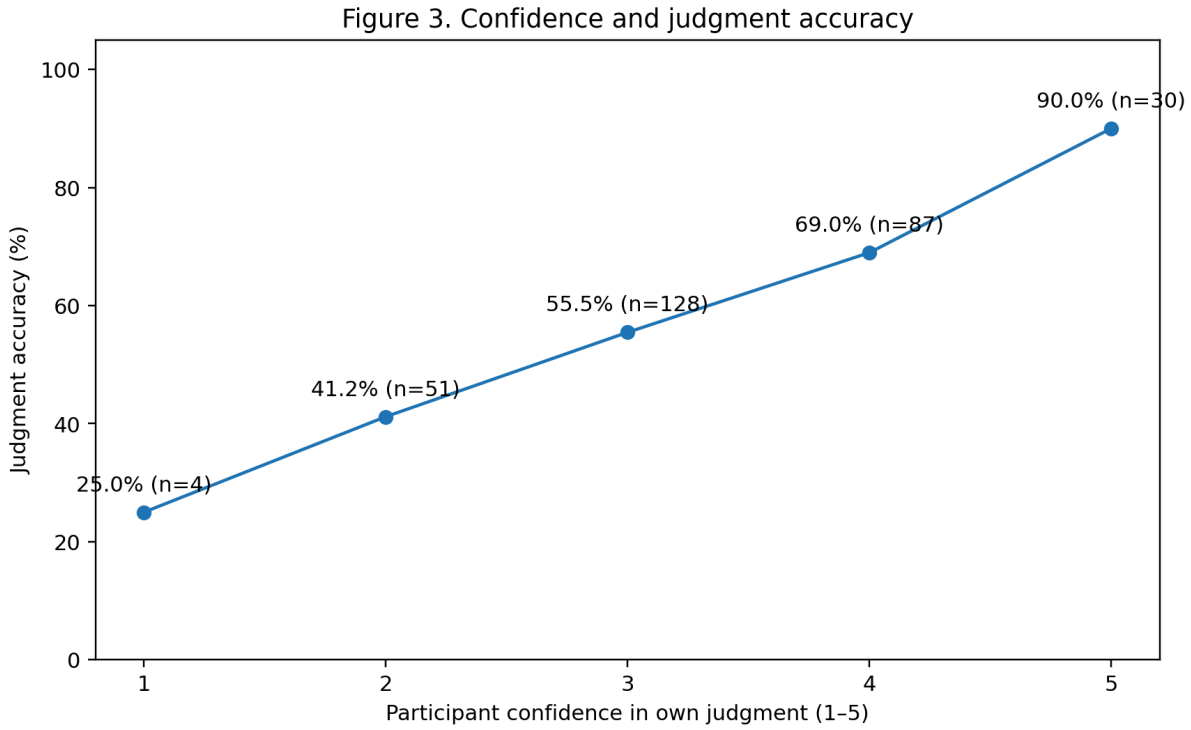


Figure 3. Participant confidence compared with judgment accuracy.

Participants reported an average confidence of **3.51/5** for correct judgments and **2.98/5** for incorrect judgments.

The difference suggests that participant trust generally corresponded to actual answer accuracy. However, the distinction was not perfect, as some incorrect answers still received relatively high trust ratings.

8.4 Presentation Conditions

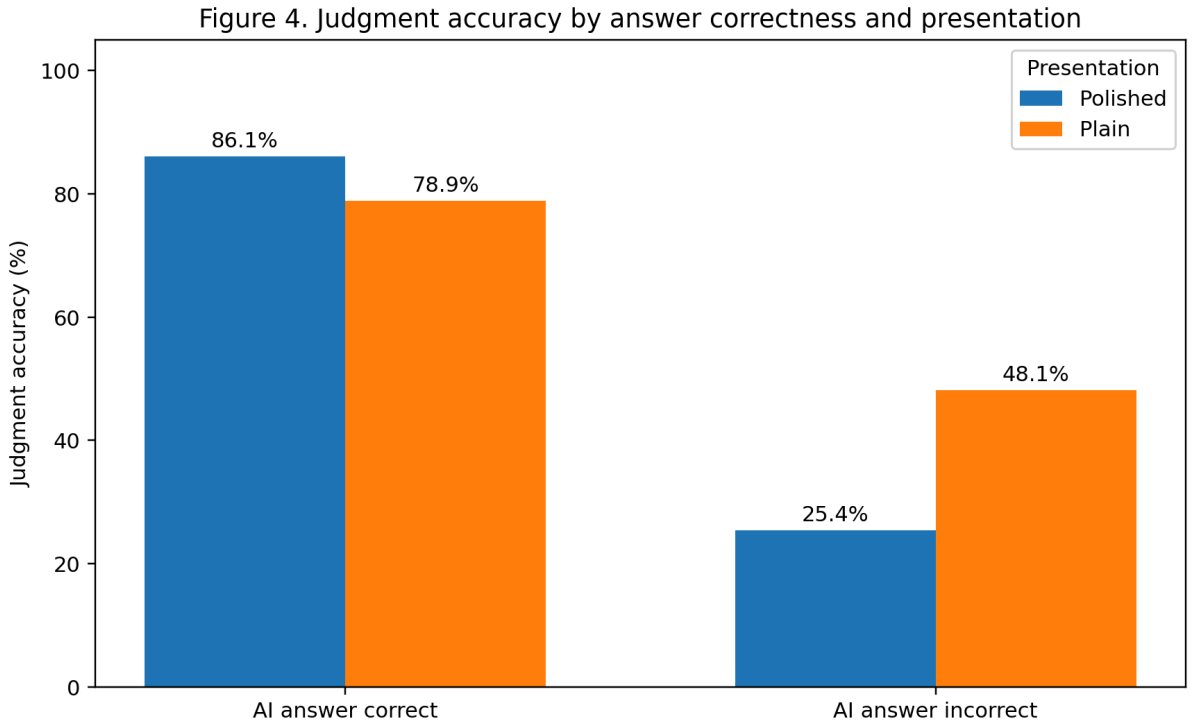


Figure 4. Judgment accuracy across presentation conditions.

Judgment accuracy ranged from **57.3%** to **62.7%** across the different presentation conditions.

The condition with **plain presentation** produced the highest observed accuracy, while **polished presentation** produced the lowest.

However, the effect differed depending on the actual correctness of the AI answer. For correct answers, polished presentation produced higher judgment accuracy **(86.1%)** than plain presentation **(78.9%)**. For incorrect answers, polished presentation produced lower judgment accuracy **(25.4%)** than plain presentation **(48.1%).**

Because the variables were not completely isolated, these differences should be interpreted as exploratory patterns rather than evidence of a direct causal effect.

8.5 Detection of Incorrect Answers

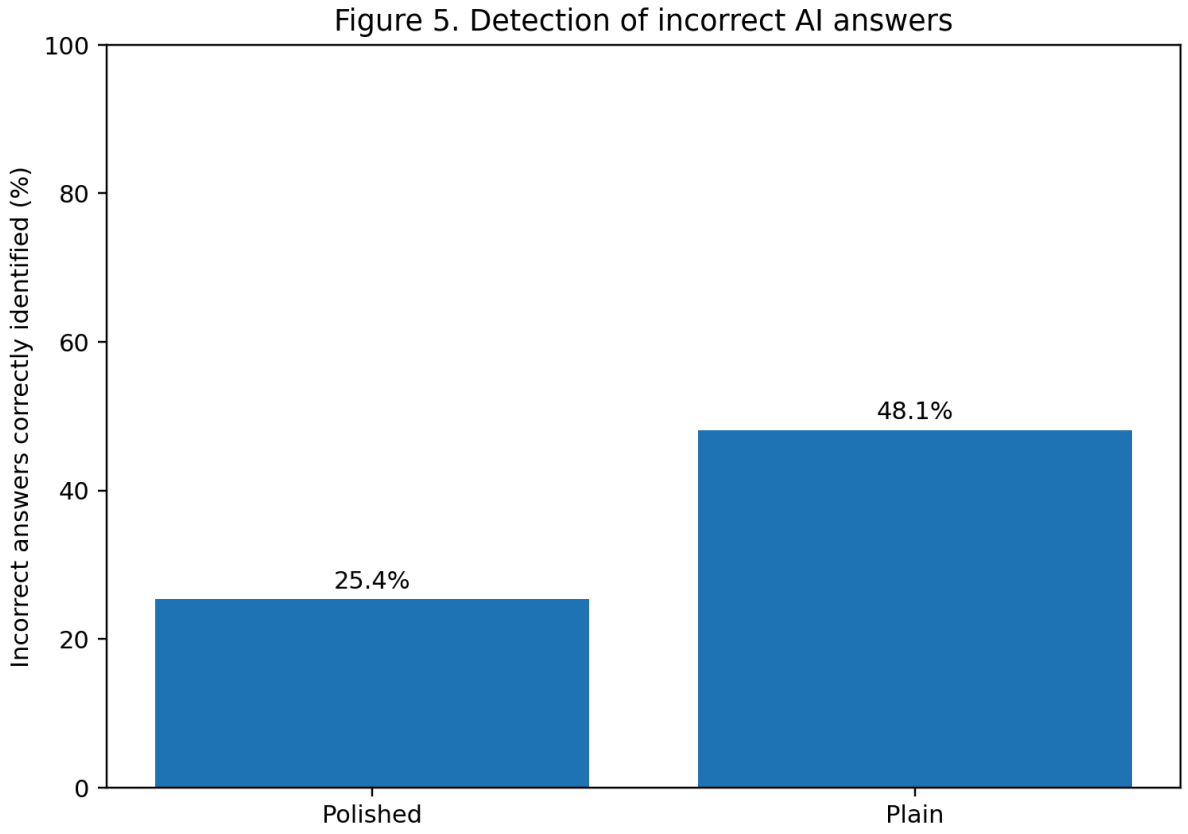


Figure 5. Participant detection of incorrect AI-generated answers.

Participants correctly identified **37.3%** of the intentionally incorrect AI answers.

This result provides an indication of how effectively participants were able to detect unreliable information.

8.6 Trust Calibration

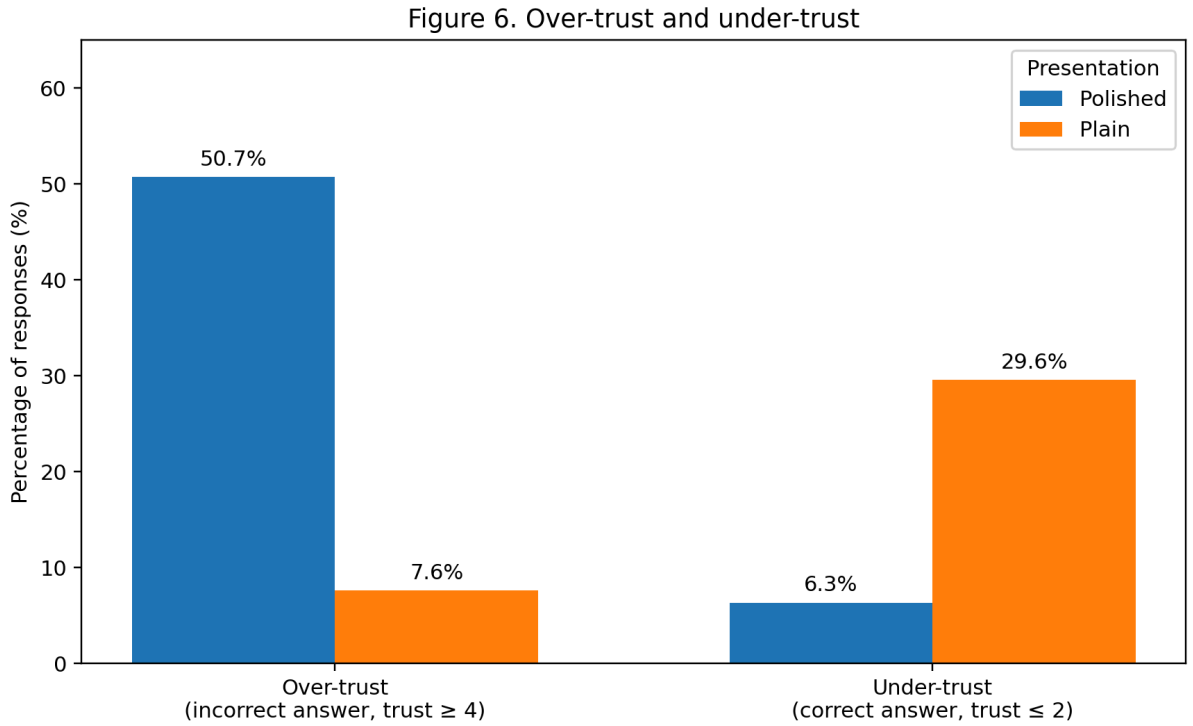


Figure 6. Relationship between participant trust and actual AI answer correctness.

The results showed that participants generally assigned higher trust ratings to correct answers than to incorrect answers. However, presentation appeared to influence the level of trust assigned

In particular, **28.0%** of incorrect answers received high trust ratings, while **17.3%** of correct answers received low trust ratings.

These patterns provide an initial indication of potential over-trust and under-trust within the sample.

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9. DISCUSSION

The main purpose of this study was to investigate whether the way AI-generated information is presented affects human trust and judgment.

The overall results showed that participants correctly judged 60.0% of the responses. Although participants generally assigned higher trust ratings to correct answers, their ability to distinguish between correct and incorrect information was not consistently reliable.

One important observation was that polished presentation was associated with higher trust, even when the AI-generated answer was incorrect. Incorrect answers presented in a polished format received an average trust rating of **3.49/5**, compared with **2.28/5** for plain answers. . This is relevant to trust calibration because users need to distinguish between information that is genuinely reliable and information that simply appears reliable.

The results showed that participants found it harder to identify incorrect answers when they were presented in a polished format. Accuracy dropped from **48.1%** with plain presentation to **25.4%** with polished presentation. This suggests that a convincing presentation may make incorrect information seem more believable.

Participants were also more confident when their judgments were correct **(3.51/5)** than when they were incorrect **(2.98/5)**. This shows that confidence generally reflected accuracy, although not always.

Overall, the findings suggest that polished presentation can increase trust in incorrect AI answers and make errors harder to recognise. However, since the data was simulated, these results cannot confirm how people would behave in real situations.

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10. COMPARISON WITH PREVIOUS RESEARCH

The findings can be considered alongside previous research into human evaluation of AI-generated information.

Steyvers et al. (2025) investigated how people assess the accuracy of large language model answers and demonstrated differences between actual model performance and human perceptions of model knowledge. The current study similarly examines whether human judgments correspond with actual answer correctness.

Kim et al. (2024) examined the effect of uncertainty expressions on user reliance and trust. The current study also considers how information presented alongside AI answers can affect human responses.

Research by Cheung and Ho (2025) examined how explainable AI can influence human trust. The current findings can therefore be considered in relation to the possibility that explanations influence perceived reliability.

The present study **partially supports** these previous findings.

However, the small sample size and exploratory design mean that the results should be interpreted cautiously and should not be treated as general conclusions about AI users.

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11. LIMITATIONS

The study has several important limitations.

Small Sample Size

Only ten participants took part in the study. This limits the statistical strength and generalisability of the findings.

Convenience Sampling

Participants were selected based on accessibility rather than random sampling. Their experiences may therefore differ from those of the wider population.

Limited Number of Questions

The experiment contains only ten questions. A larger question set would provide more observations and reduce the influence of individual questions.

Variables Were Not Fully Isolated

Confidence signals, explanations, and answer correctness were presented in different combinations. As a result, the experiment cannot determine the independent effect of each individual variable.

Prior Knowledge

Participants may have already known some answers. This could affect their judgments independently of how the AI answer was presented.

Self-Reported Trust

Trust and intended reliance were measured through participant responses. These measures may not always reflect how participants would behave in a real situation.

Despite these limitations, the study is useful as a pilot investigation and provides a foundation for a more controlled future experiment.

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12. ETHICAL CONSIDERATIONS

Participant privacy was considered throughout the study.

Participants were identified using anonymous participant IDs rather than names, and unnecessary personal information was not collected.

Participation is voluntary, and participants should understand the general purpose of the experiment.

Because some answers were intentionally incorrect, participants receive an appropriate debrief after completing the experiment.

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13. FUTURE WORK

The experiment could be expanded in several ways.

First, a larger participant sample would provide a stronger dataset and allow more reliable comparisons.

Second, the number of questions could be increased and balanced across different subject areas.

Third, future research could use a fully controlled factorial design. Confidence, explanations, evidence quality, and uncertainty could each be manipulated independently while keeping the underlying answer unchanged.

Future studies could also investigate the quality of supporting evidence. For example, an answer could be presented with no evidence, weak evidence, strong evidence, or conflicting evidence.

Another possible extension would be to measure participants' previous experience with AI. This could determine whether frequent AI users evaluate AI-generated information differently.

Finally, future experiments could measure actual behaviour rather than relying only on self-reported trust. Participants could be given the choice to accept an AI answer, verify it, or reject it, providing a more direct measure of reliance.

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14. CONCLUSION

This study investigated how people evaluate AI-generated information and whether their trust corresponds to the actual accuracy of AI answers.

An interactive Python and Streamlit platform was developed to conduct the experiment and collect anonymous participant responses. Thirty participants evaluated ten AI-generated answers and reported their perceived correctness, confidence, trust, and intended reliance.

The main finding was that polished presentation increased trust, even when the answers were incorrect. Participants also found it harder to identify incorrect answers when they were presented in a polished format.

The study also found that participants were generally more confident when their judgments were correct than when they were incorrect, suggesting a relationship between confidence and judgment accuracy.

The hypothesis that presentation cues may influence trust was **partially supported** by the observed results.

Although the study is limited by its small sample and exploratory design, it demonstrates an important issue in human-AI interaction. The challenge is not simply whether people trust AI, but whether they know **when that trust is justified**.

As AI-generated information becomes increasingly common, users need to evaluate answers based on evidence and accuracy rather than relying solely on confidence, explanations, or persuasive presentation.

This project provides an initial investigation into that problem while demonstrating the use of software development, experimental research, data collection, and statistical analysis in a single interdisciplinary project.

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SUPPLEMENTARY MATERIALS

Research materials, experimental code, questionnaire, and supporting files are available at:

https://github.com/celineajapardi-netizen/ai-trust-calibration-research